

Frequency Response

Figure 21.20 shows the AC frequency response circuit for the CS amplifier with current source load in Fig. 21.17. V_{DD} in this figure is drawn as an AC ground. The source-gate capacitance of M2 isn't shown because both sides of it are at AC ground (DC voltages), so it doesn't affect the frequency behavior of the circuit. The pole associated with the input node of the circuit is

$$f_{in} = \frac{1}{2\pi(C_{gs1} + C_{gd1}(1 + |A_v|)) \cdot R_s} \quad (21.41)$$

while the pole associated with the output node of the circuit is located at

$$f_{out} = \frac{1}{2\pi\left(C_{dg2} + C_{gd1}\left(1 + \frac{1}{|A_v|}\right)\right) \cdot r_{o1} \parallel r_{o2}} \quad (21.42)$$

where $A_v = -g_{m1} \cdot r_{o1} \parallel r_{o2}$ (see Eq. [21.39]). Using Eq. (21.42) to calculate the output pole, results in a *wrong estimate* when A_v is large. As we'll see in a moment, the fact that A_v decreases above f_{in} causes v_{in} to “flatten out” because the Miller capacitance, $C_{gd1}(1 + |A_v|)$, gets smaller. The effect of v_{in} ceasing to decrease causes the output pole to appear much larger than what is predicted by Eq. (21.42). This important phenomenon is called *pole splitting*.

The zero in the transfer function (see Eq. (21.15)) is located at

$$f_z = \frac{g_{m1}}{2\pi C_{gd1}} \quad (21.43)$$

The transfer function of the amplifier is then estimated as

$$A_v(f) = -g_{m1} \cdot (r_{o1} \parallel r_{o2}) \cdot \frac{\left(1 - j\frac{f}{f_z}\right)}{\left(1 + j\frac{f}{f_{in}}\right)\left(1 + j\frac{f}{f_{out}}\right)} \quad (21.44)$$

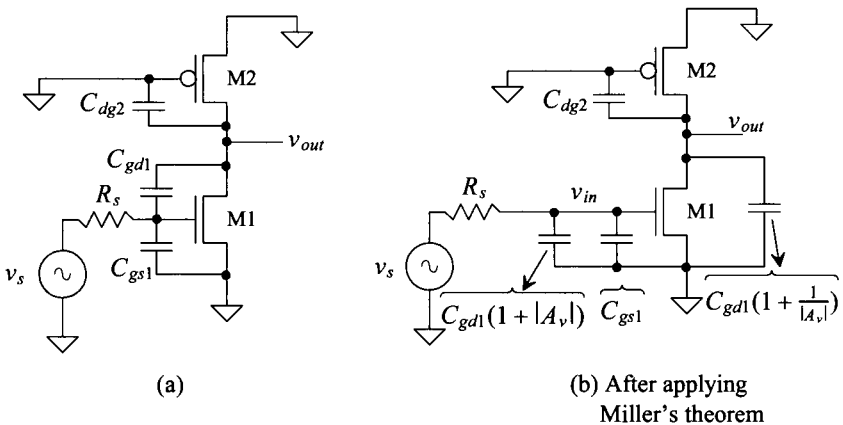


Figure 21.20 Frequency response of the CS amplifier with current source load.

We can then write the magnitude as

$$|A_v(f)| = \frac{g_{m1} \cdot (r_{o1} || r_{o2}) \cdot \sqrt{1 + \left(\frac{f}{f_z}\right)^2}}{\left(\sqrt{1 + \left(\frac{f}{f_{in}}\right)^2}\right) \cdot \left(\sqrt{1 + \left(\frac{f}{f_{out}}\right)^2}\right)} \quad (21.45)$$

and the phase shift through the amplifier as

$$\angle A_v = 180 - \tan^{-1}\left(\frac{f}{f_z}\right) - \tan^{-1}\left(\frac{f}{f_{in}}\right) - \tan^{-1}\left(\frac{f}{f_{out}}\right) \quad (21.46)$$

Example 21.6

Determine the gain, magnitude, and phase shift of the amplifier shown in Fig. 21.21. Using an AC SPICE simulation, verify your hand calculations.

For the AC small-signal analysis, the big resistor appears like an open and the big capacitor a short. These components are added to ensure that M1 is biased to sink the current supplied by M2.

The low-frequency gain of the circuit is, from Eq. (21.39) and Table 9.1,

$$|A_v| = (150 \mu A/V) \cdot (5M\Omega || 4M\Omega) = 333 V/V \rightarrow 50 \text{ dB}$$

The input capacitance is

$$C_{in} = (C_{gs1} + C_{gd1}(1 + |A_v|)) = (23.3 + 2 \cdot 333) \text{ fF} = 691 \text{ fF}! \quad (21.47)$$

The output capacitance is

$$C_{out} = \left(C_{dg2} + C_{gd1}\left(1 + \frac{1}{|A_v|}\right)\right) \approx 6 + 2 \text{ fF} = 8 \text{ fF} \quad (21.48)$$

Using Eqs. (21.41)–(21.43), we get $f_{in} = 2.3 \text{ MHz}$, $f_{out} = 9 \text{ MHz}$, and $f_z = 11.9 \text{ GHz}$. Figure 21.22 shows the simulation results. As the discussion following Eq. (21.42) indicated, the estimate for f_{out} is wrong. To determine the correct value, let's discuss pole splitting (see Eq. [21.62]). ■

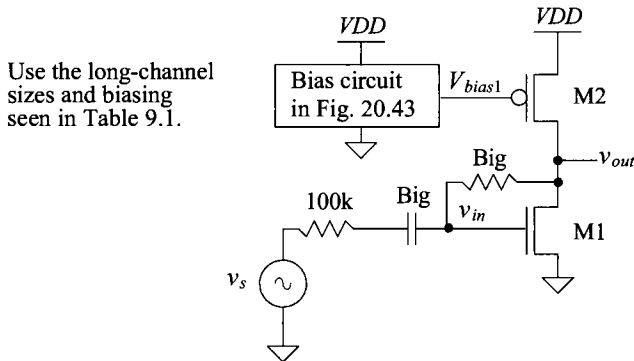


Figure 21.21 Amplifier used in Ex. 21.6.

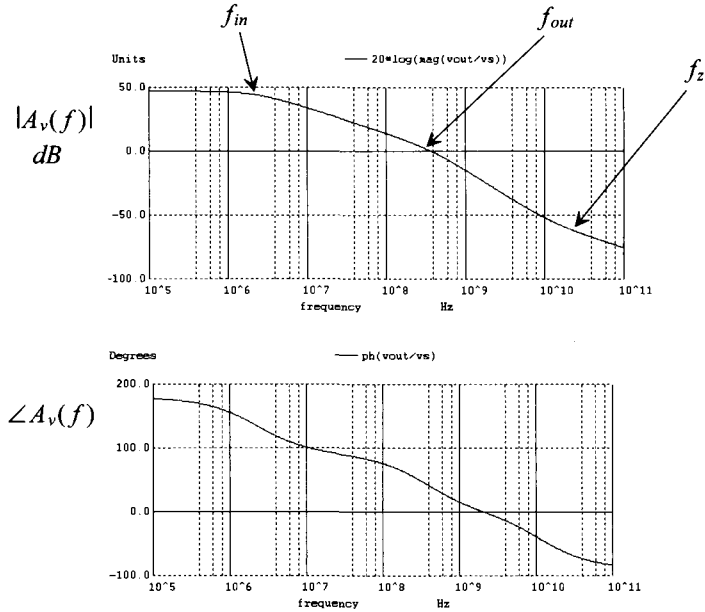


Figure 21.22 Frequency response of the amplifier in Fig. 21.21.

Pole Splitting

As discussed after Eq. (21.42), using Miller’s theorem to calculate the location of the output pole when the gain of the amplifier is large results in error. Clearly, as just demonstrated in Ex. 21.6, the calculated value of f_{out} is 9 MHz while the SPICE simulation in Fig. 21.22 shows that f_{out} is approximately 200 MHz. The source of this discrepancy can be traced to the fact that above f_{in} the gain of the amplifier, A_v , decreases causing the effective input capacitance of the amplifier to decrease. As seen in Fig. 21.23, this causes the input voltage to “flatten out” instead of continuing to fall at -20 dB/decade. The result is the second, or output pole, appears to split from (leave) the first,

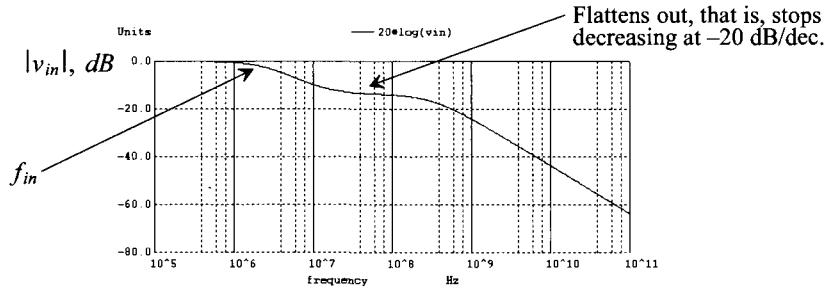


Figure 21.23 Showing how the amplifier’s, in Fig. 21.21, input voltage flattens out as the gain decreases.

or input pole, and move to a higher frequency. Note that if we resimulate the amplifier in Fig. 21.21 with the source resistance set to zero ohms (so there isn't an input pole), pole splitting isn't present and Eq. (21.42) can be used to calculate the location of the output pole.

To characterize this effect, let's use the circuit in Fig. 21.7 with the addition of a source resistance, R_s , as seen in Fig. 21.24. In this figure, $R_o = r_{o1} || r_{o2}$ and $C_o = C_{dg2}$. From Eq. (21.14), we can write

$$\frac{v_{out}}{v_{in}} = -g_{m1} R_o \cdot \frac{1 - j\omega \frac{C_{gd1}}{g_{m1}}}{1 + j\omega(C_{gd1} + C_o) \cdot R_o} \quad (21.49)$$

At the gate of M1, we can write

$$\frac{v_{in} - v_s}{R_s} + \frac{v_{in}}{1/j\omega C_{gs1}} + \frac{v_{in} - v_{out}}{1/j\omega C_{gd1}} = 0 \quad (21.50)$$

or

$$v_{in} = \frac{\frac{v_s}{R_s} + v_{out} \cdot j\omega C_{gd1}}{\frac{1}{R_s} + j\omega C_{gs1} + j\omega C_{gd1}} \quad (21.51)$$

Substituting Eq. (21.51) into Eq. (21.49) gives

$$v_{out} = -g_{m1} R_o \cdot \frac{1 - j\omega \frac{C_{gd1}}{g_{m1}}}{1 + j\omega(C_{gd1} + C_o) \cdot R_o} \cdot \frac{\frac{v_s}{R_s} + v_{out} \cdot j\omega C_{gd1}}{\frac{1}{R_s} + j\omega C_{gs1} + j\omega C_{gd1}} \quad (21.52)$$

or, with $s = j\omega$,

$$\frac{v_{out}}{v_s} = \frac{-g_{m1} R_o \cdot \left(1 - s \frac{C_{gd1}}{g_{m1}}\right)}{s^2 [R_o R_s (C_{gd1} C_{gs1} + C_o C_{gs1} + C_o C_{gd1})] + s [(C_{gd1} + C_o) R_o + (C_{gs1} + C_{gd1}) R_s + C_{gd1} g_{m1} R_o R_s] + 1} \quad (21.53)$$

The zero is still, as indicated in Eqs. (21.15) and (21.43), located at

$$f_z = \frac{g_{m1}}{2\pi C_{gd1}} \quad (21.54)$$

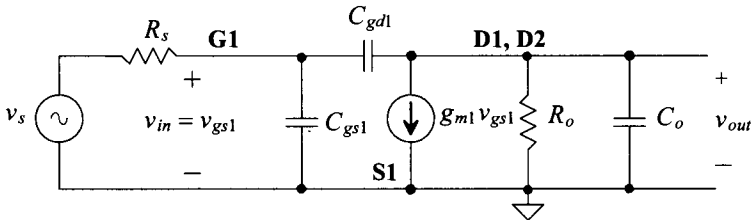


Figure 21.24 Circuit used to calculate the transfer function of the amplifier in Fig. 21.21.

At low frequencies the denominator is approximately (the coefficient of the s^2 term is small)

$$1 + j\omega \cdot [(C_{gd1} + C_o)R_o + (C_{gs1} + C_{gd1})R_s + C_{gd1}g_{m1}R_oR_s] \quad (21.55)$$

and so the low-frequency pole, using $|A_v| = g_{m1}(r_{o1} || r_{o2}) = g_{m1}R_o$, $C_o = C_{dg2}$, is located at

$$f_1 \approx \frac{1}{2\pi[(C_{gd1} + C_{dg2}) \cdot r_{o1} || r_{o2} + (C_{gs1} + C_{gd1}(1 + |A_v|)) \cdot R_s]} \quad (21.56)$$

If $C_{gd1}(1 + |A_v|)$ is much larger than the other capacitances,

$$f_1 \approx \frac{1}{2\pi C_{gd1}(1 + |A_v|) \cdot R_s} \quad (21.57)$$

Notice the similarity to Eq. (21.41), where the pole is associated with the input of the amplifier. Further notice that if we had used this result in Ex. 21.6, the input pole would have changed very little. Also notice that if $R_s = 0$, then

$$f_1 \approx \frac{1}{2\pi(C_{gd1} + C_{dg2}) \cdot r_{o1} || r_{o2}} \quad (21.58)$$

Notice the similarity to Eq. (21.42), where the pole is associated with the output of the amplifier. To determine the location of the second root of the denominator (the second pole location, f_2), let's factor Eq. (21.55) from the denominator of Eq. (21.53) or

$$[1 + s[(C_{gd1} + C_o)R_o + (C_{gs1} + C_{gd1})R_s + C_{gd1}g_{m1}R_oR_s]] \cdot \left(1 + \frac{s^2[R_oR_s(C_{gd1}C_{gs1} + C_oC_{gs1} + C_oC_{gd1})]}{s[(C_{gd1} + C_o)R_o + (C_{gs1} + C_{gd1})R_s + C_{gd1}g_{m1}R_oR_s] + 1} \right) \quad (21.59)$$

where this equation is in the form

$$\left(1 + j \cdot \frac{f}{f_1} \right) \cdot \left(1 + j \cdot \frac{f}{f_2} \right) \quad (21.60)$$

Note again that if $R_s = 0$, the second term in Eq. (21.59) goes to unity. Looking at the second term of Eq. (21.59), let's divide the numerator and denominator by sR_oR_s

$$1 + j \cdot \frac{2\pi f \cdot (C_{gd1}C_{gs1} + C_oC_{gs1} + C_oC_{gd1})}{(C_{gd1} + C_o)/R_s + (C_{gs1} + C_{gd1})/R_o + C_{gd1}g_{m1} + 1/sR_oR_s} \quad (21.61)$$

In any practical MOSFET amplifier, $g_m \gg 1/r_o$ (if not, then the magnitude of the open circuit gain, Eq. (21.40), is too small for the MOSFET to be of practical value as an amplifying device). Also, the R_s is assumed to be large (if not, then f_2 is not used), that is, the amplifier's transfer function has a single-pole response, where the pole is associated with the output of the amplifier as seen in Eq. (21.58). We can therefore write

$$f_2 \approx \frac{g_{m1}C_{gd1}}{2\pi \cdot (C_{gd1}C_{gs1} + C_oC_{gs1} + C_oC_{gd1})} \quad (21.62)$$

Calculating the location of the second pole in Ex. 21.6 with this equation results in $f_2 = 240$ MHz (which is very close to the simulated location).

Why the name pole splitting? If we increase the effective size of C_{gd1} by placing a capacitor, C_c , from the amplifier's input to its output (so that C_c is in parallel with C_{gd1} , that is, the effective value is $C_c + C_{gd1}$), then, as Eq. (21.56) shows, the low-frequency location of the pole, f_1 , decreases. At the same time, increasing the effective size of C_{gd1} causes the location of the high-frequency pole f_2 , as seen in Eq. (21.62), to increase. Thus the name *pole splitting*.

Pole Splitting Summary

Because of the ubiquity of pole splitting in CMOS amplifier design, let's summarize our discussion using a generic model, Fig. 21.25. In terms of the parameters seen in this figure the location of the zero is now

$$f_z = \frac{g_{m2}}{2\pi \cdot C_c} \quad (21.63)$$

The first pole is located at

$$f_1 \approx \frac{1}{2\pi[(C_c + C_2) \cdot R_2 + (C_1 + C_c(1 + g_{m2}R_2)) \cdot R_1]} \quad (21.64)$$

For large gain, $g_{m2}R_2$, we can simplify this to

$$f_1 \approx \frac{1}{2\pi g_{m2}R_2R_1C_c} \quad (21.65)$$

The second pole is located at

$$f_2 \approx \frac{g_{m2}C_c}{2\pi \cdot (C_cC_1 + C_1C_2 + C_cC_2)} \quad (21.66)$$

The transfer function is then written as

$$A_v(f) = \frac{v_{out}(f)}{v_s(f)} = g_{m1}R_1g_{m2}R_2 \cdot \frac{\left(1 - j \cdot \frac{f}{f_z}\right)}{\left(1 + j \cdot \frac{f}{f_1}\right) \cdot \left(1 + j \cdot \frac{f}{f_2}\right)} \quad (21.67)$$

where $A_v = g_{m1}R_1g_{m2}R_2$ (the gain is now positive because of the defined direction of the current source in Fig. 21.25).

Again note that if C_c is increased, f_1 moves downwards and f_2 moves upwards (pole splitting). Also note that if the capacitor, C_c , is made large (quite common), then f_1 is much lower than the location of the other pole or the zero (the pole associated with f_1 is

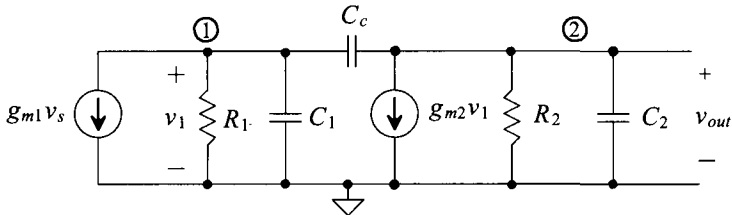


Figure 21.25 Generic model used to estimate bandwidth in a CMOS amplifier.

said to be the *dominant pole*). The amplifier's transfer function, with the help of Eq. (21.65), can be written as

$$A_v(f) = \frac{v_{out}(f)}{v_s(f)} \approx \frac{g_{m1}R_1g_{m2}R_2}{\left(1+j\cdot\frac{f}{f_1}\right)} = \frac{g_{m1}R_1g_{m2}R_2}{(1+j\cdot 2\pi f\cdot g_{m2}R_2R_1C_c)} \quad (21.68)$$

This equation is in the form

$$A_v(f) = \frac{v_{out}(f)}{v_s(f)} = \frac{A_{DC}}{1+j\cdot\frac{f}{f_{3dB}}} \quad (21.69)$$

where

$$A_{DC} = g_{m1}R_1g_{m2}R_2 \text{ and } f_{3dB} = \frac{1}{2\pi g_{m2}R_2R_1C_c} \quad (21.70)$$

The magnitude and phase responses of a generic CMOS amplifier are seen in Fig. 21.26. Note that at frequencies much larger than the amplifier's 3 dB frequency, that is, $f/f_{3dB} \gg 1$, Eq. (21.68) can be written as

$$A_v(f) = \left| \frac{v_{out}(f)}{v_s(f)} \right| \approx \frac{g_{m1}}{2\pi f\cdot C_c} \quad (21.71)$$

To determine the frequency, f_{un} , where the transfer function is unity, we set Eq. (21.71) to one and solve to yield

$$f_{un} = \frac{g_{m1}}{2\pi C_c} \quad (21.72)$$

We've derived some very useful equations, so let's apply them in some examples.

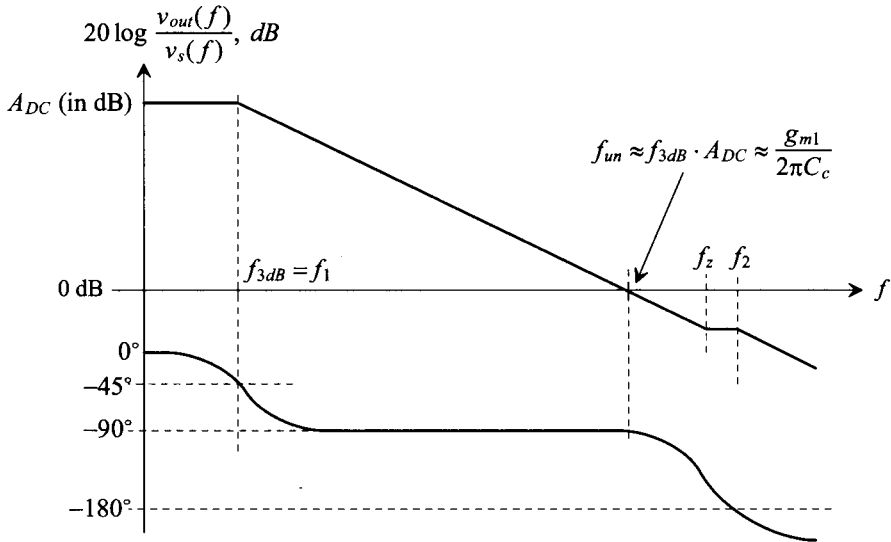


Figure 21.26 Magnitude and phase responses of a generic CMOS amplifier.

Example 21.7

Repeat Ex. 21.6 using the model in Fig. 21.25.

Using the model parameters in Fig. 21.25 and looking at Figs. 21.20a and 21.21, we can write (with the help of Table 9.1)

$$R_1 = R_s = 100\text{k}, C_1 = C_{gs1} = 23.3\text{ fF}, C_c = C_{gd1} = 2\text{ fF},$$

$$C_2 = C_{dg2} = 6\text{ fF}, R_2 = r_{o1} \parallel r_{o2} = 2.22\text{ M}\Omega$$

$$g_{m2} \text{ (in Fig. 21.25)} = g_{m1} \text{ (in Fig. 21.21)} = 150\text{ }\mu\text{A/V}$$

$$g_{m1} \text{ (in Fig. 21.25)} = 1/R_s \text{ (in Fig. 21.21)} = 10\text{ }\mu\text{A/V}$$

Using Eq. (21.67) or (21.70), we have $|A_v| = 333\text{ V/V} \rightarrow 50\text{ dB}$. Using Eq. (21.63), we get $f_z = 11.9\text{ GHz}$; Eq. (21.65), we get $f_1 = 2.3\text{ MHz}$; and finally using Eq. (21.66), we get $f_2 = 240\text{ MHz}$. These calculations should be compared to the simulation results seen in Fig. 21.22. ■

Example 21.8

Repeat Ex. 21.7 if R_s is zero. Verify the hand calculations using SPICE.

Again we use the model seen in Fig. 21.25. From Ex. 21.7, $g_{m1}R_1$ is unity (and so R_s doesn't affect the gain). Further, the location of the zero doesn't change. Since R_s is zero ($R_1 = 0$ so the current source, $g_{m1}v_s$, and the resistor, R_1 in Fig. 21.25 are replaced with a voltage source, v_s), the location of f_2 is moved to an infinite frequency (as seen in Eq. [21.59]). Using Eq. (21.64), we get

$$f_1 \approx \frac{1}{2\pi(C_{gd1} + C_{dg2}) \cdot r_{o1} \parallel r_{o2}} = \frac{1}{2\pi(8\text{ fF}) \cdot 2.22\text{ M}} = 9\text{ MHz}$$

The simulation results are seen in Fig. 21.27. Note the small perturbation in the frequency response above f_1 . The response is not perfectly decreasing at -20 dB/decade . This imperfection can be traced to the bias voltage, V_{bias1} , used on the gate of M2. It is not a perfect AC ground as was assumed in our hand calculations. By placing a capacitor on the gate of M2 to ground, the variation in V_{bias1} is reduced and the amplifier behaves as expected. This is an *important point*. It is often a good idea to “bypass” the bias voltages to AC ground to reduce their impedance in practical design. This also keeps circuits that are biased from the same bias circuit from interacting. ■

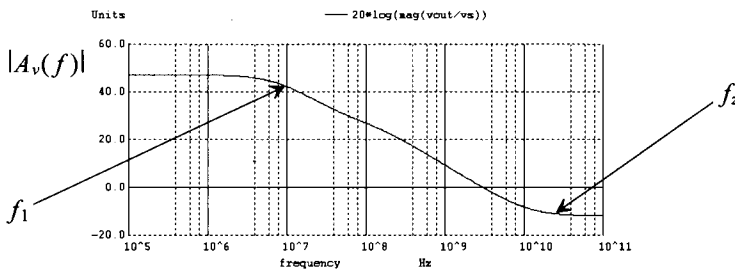


Figure 21.27 Simulation results for Ex. 21.8.

Example 21.9

Determine the frequency response of the amplifier seen in Fig. 21.28. Verify the hand calculations using simulations.

Use the long-channel sizes and biasing seen in Table 9.1.

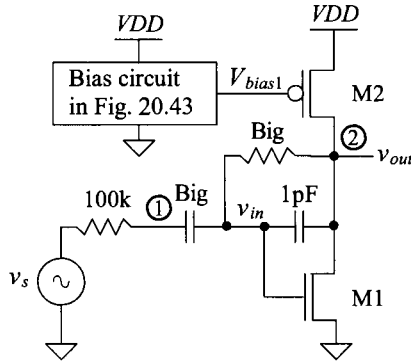


Figure 21.28 Amplifier used in Ex. 21.9.

The only thing that is different between this amplifier and the one in Ex. 21.7 is the value of C_c . In Fig. 21.28

$$C_c = 1 \text{ pF} + C_{gd1} \approx 1 \text{ pF}$$

We can then write, using Eq. (21.65)

$$f_1 \approx \frac{1}{2\pi \cdot (150 \mu\text{A/V}) \cdot 100\text{k} \cdot 2.2\text{MEG} \cdot 1\text{pF}} = 4.8 \text{ kHz}$$

using Eq. (21.66)

$$f_2 \approx \frac{(150 \mu\text{A/V}) \cdot 1\text{p}}{2\pi \cdot (1\text{p} \cdot 23.3\text{f} + 23.3\text{f} \cdot 6\text{f} + 1\text{p} \cdot 6\text{f})} = 811 \text{ MHz}$$

At the risk of stating the obvious, these two poles are split apart by a great distance with the addition of the 1 pF capacitor. The zero, from Eq. (21.63), is located at

$$f_z = \frac{150 \mu\text{A/V}}{2\pi \cdot 1\text{p}} = 23 \text{ MHz}$$

The unity-gain frequency is found using Eq. (21.72) with $R_s = 1/g_{m1}$.

$$f_{un} = \frac{1}{2\pi R_s C_c} = 1.59 \text{ MHz}$$

The simulation results are seen in Fig. 21.29. The value of the gain is 47 dB (we calculated 50 dB). The value of the first pole is approximately 6 kHz (we calculated 4.8 kHz). The small discrepancies are most likely the result of differences in the actual circuit parameters compared to the values we used in the formulas. ■

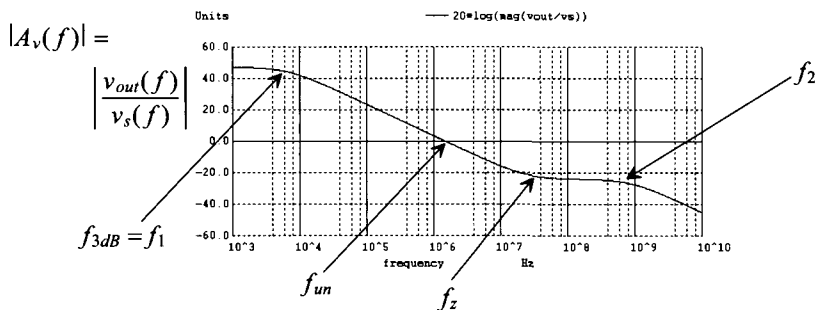


Figure 21.29 Frequency response of the amplifier in Fig. 21.28.

Example 21.10

Estimate the frequency response of the amplifier seen in Fig. 21.30. Verify the estimates with SPICE.

MA's gate and drain, at DC, are at the same potential. This biases MB to sink the right amount of current from MD.

Using the model in Fig. 21.25, we see that the source voltage, v_s , modulates the drain current in MA. We can therefore write

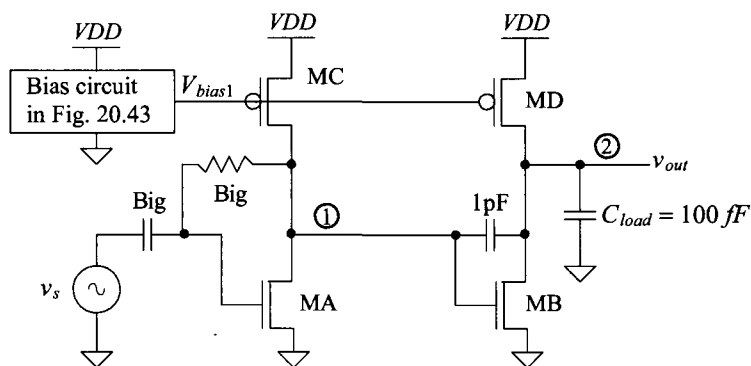
$$g_{mA}v_s = g_{m1}v_s \rightarrow g_{m1} \text{ (in Fig. 21.25)} = g_{mA} \text{ (in Fig. 21.30)} = 150 \mu A/V$$

$$g_{m2} \text{ (in Fig. 21.25)} = g_{mB} \text{ (in Fig. 21.30)} = 150 \mu\text{A/V}$$

$$R_1 = r_{oA} || r_{oC} = R_2 = r_{oB} || r_{oD} = 2.22 \text{ M}\Omega$$

To determine the capacitance on nodes 1 and 2, we can use Fig. 21.31. At node 1

$$C_1 = C_{gdA} + C_{dgC} + C_{gsB} = 31.3 \text{ fF}$$



Use the long-channel sizes and biasing seen in Table 9.1.

Figure 21.30 Amplifier used in Ex. 21.10.

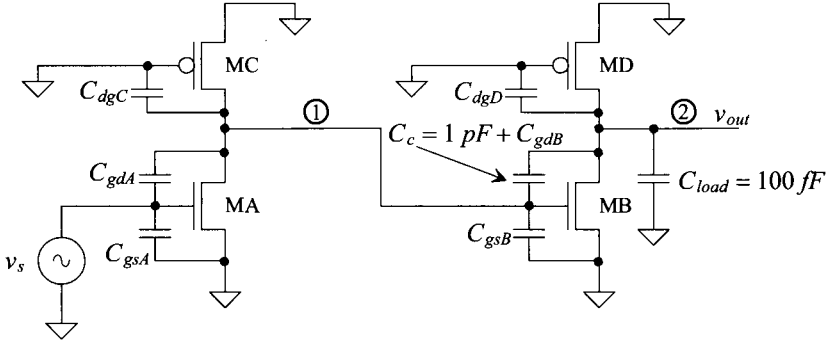


Figure 21.31 The capacitances for the amplifier in Fig. 21.30.

where C_{gdA} is the output Miller capacitance of MA ($C_{gdA} \approx C_{gdA} \left(1 + \frac{1}{|A|}\right)$ when the gain of MA is large). At node 2

$$C_2 = C_{load} + C_{dgD} = 106 \text{ fF}$$

The value of C_c is

$$C_c = 1 \text{ pF} + C_{gdB} \approx 1 \text{ pF} \text{ (just the added capacitor)}$$

The gain of the topology is estimated as

$$A_{DC} = g_{m1} R_1 g_{m2} R_2 = (150 \mu\text{A/V})^2 (2.22 \text{ M}\Omega)^2 = 110,889 \text{ V/V} \rightarrow 101 \text{ dB}$$

The poles and zero can be calculated using Eqs. (21.63), (21.65), and (21.66)

$$f_z = \frac{150 \mu\text{A/V}}{2\pi \cdot 1.002 \text{ pF}} = 23 \text{ MHz}$$

$$f_1 \approx \frac{1}{2\pi \cdot (150 \mu\text{A/V})(2.22 \text{ M}\Omega)^2 (1.002 \text{ pF})} = 214 \text{ Hz}$$

$$f_2 \approx \frac{(150 \mu\text{A/V}) \cdot 1.002}{2\pi \cdot (1.002 \cdot 0.0313 + 0.0313 \cdot 0.106 + 1.002 \cdot 0.106) \text{ pF}} = 170 \text{ MHz}$$

The location of the unity gain frequency is calculated using Eq. (21.72) as 23 MHz. The simulation results are seen in Fig. 21.32. ■

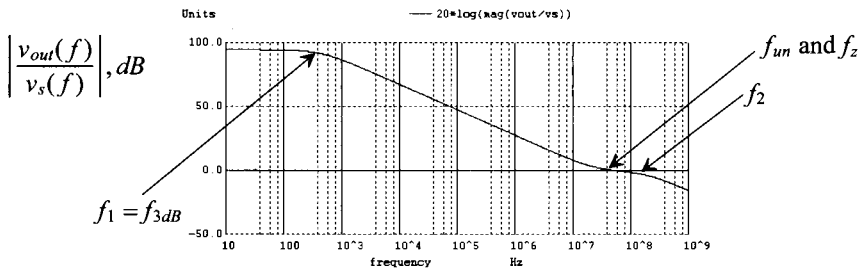


Figure 21.32 Frequency response of the amplifier in Fig. 21.30.